

North East University Bangladesh

Department of Computer Science and Engineering



An MLP Framework for Financial Transaction Fraud Detection

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Abstract

Fraud detection is essential for protecting financial institutions from economic losses and maintaining the integrity of banking systems. Detecting fraudulent transactions remains a significant challenge due to the highly imbalanced nature of transaction data, where legitimate transaction records greatly outnumber fraudulent ones. Moreover, fraudsters continuously adapt and evolve their tactics to evade existing fraud detection mechanisms, creating the need for more accurate, robust, and reliable fraud detection models.

This study presents a deep learning-based framework for detecting fraudulent bank transactions through the evaluation and comparison of multiple deep learning models using pre-processed bank transfer and cash withdrawal transaction data. The models were systematically compared to determine the most effective approach for fraud detection. Based on the experimental results, the Multilayer Perceptron (MLP) model achieved the best overall performance, obtaining a ROC-AUC score of 0.999058, a PR-AUC score of 0.996666, and a Fraud-F1 score of 0.977474.

The findings indicate that the proposed framework can effectively enhance fraud detection, maintain reliable performance on highly imbalanced data, and strengthen the security and integrity of modern financial systems.

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Chapter 1

Introduction

1.1 Background

Digital financial services have grown rapidly worldwide, making financial transactions faster, more convenient, and easily accessible. However, this growth has also increased the number of fraudulent transactions, creating significant financial losses and security concerns for financial institutions. Fraud detection is particularly challenging because fraudulent transactions are rare compared to legitimate ones and often resemble normal customer behavior. Moreover, fraudsters continuously change their techniques to evade existing detection systems.

In Bangladesh, digital banking services are widely used for money transfers, payments, and cash withdrawals. Although this study is not based on Bangladesh-specific fraud statistics, these real-world scenarios motivated us to focus on TRANSFER and CASH_OUT transactions, as they are commonly associated with fraudulent activities.

Recent studies have shown that deep learning models outperform traditional rule-based systems by learning complex transaction patterns directly from data. Motivated by these findings, this project develops a deep learning-based fraud detection framework. After pre-processing the dataset and handling class imbalance using resampling and class weighting, six deep learning models, namely MLP, GRU, BiLSTM, TabNet, FT-Transformer, and TabTransformer, were trained and evaluated using standard performance metrics.

1.2 Problem Statement

Fraudulent financial transactions remain difficult to detect because they represent only a small fraction of overall transactions, resulting in a highly imbalanced dataset. In addi-

tion, fraudsters continuously adapt their strategies, making traditional rule-based detection systems less effective. Therefore, there is a need for an intelligent and reliable fraud detection framework that can accurately identify fraudulent transactions while minimizing false alarms.

1.3 Objectives

The main objective of this project is to develop and evaluate a deep learning-based framework for detecting fraudulent financial transactions.

The specific objectives are:

- To preprocess financial transaction data and perform feature engineering.
- To handle class imbalance using appropriate techniques.
- To develop and compare six deep learning models for fraud detection.
- To identify the best-performing model using standard evaluation metrics.

1.4 Scope of the Project

This project focuses on detecting fraudulent `TRANSFER` and `CASH_OUT` transactions using deep learning techniques. The work includes data preprocessing, feature engineering, handling class imbalance, model training, and performance evaluation. The models are compared using metrics such as Precision, Recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix. The project does not include real-time deployment or integration with live banking systems.

Chapter 2

Dataset & Methodology

2.1 Dataset

This study uses the PaySim dataset for binary fraud detection, where the target variable, `isFraud`, represents fraudulent (1) and legitimate (0) transactions. Although the dataset contains multiple transaction types, only the `TRANSFER` and `CASH_OUT` transaction types were selected because they are directly associated with fraudulent financial activities.

The dataset contains transaction information such as transaction type, transaction amount, transaction step, and the sender's and receiver's account balances before and after each transaction. These features provide useful information for identifying abnormal transaction patterns.

step	type	amount	nameOrig	oldbalanceOrig	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
164	TRANSFER	494,531.92	C1085858428	0.00	0.00	C1289313812	1,149,485.91	1,644,017.84	0	0
282	CASH_OUT	25,494.83	C1689585489	0.00	0.00	C283972309	178,520.93	204,015.76	0	0
576	CASH_OUT	1,322,277.84	C1585759405	1,322,277.84	0.00	C1855735331	0.00	1,322,277.84	1	0
466	TRANSFER	104,171.37	C952245558	104,171.37	0.00	C2101227966	0.00	0.00	1	0
283	CASH_OUT	212,105.20	C1446494202	446.00	0.00	C460025153	144,622.75	356,727.96	0	0
249	CASH_OUT	134,228.50	C471329919	0.00	0.00	C2121315520	295,869.35	430,097.85	0	0
436	CASH_OUT	539,658.73	C1985701449	539,658.73	0.00	C1862889538	0.00	539,658.73	1	0
564	TRANSFER	2,472,250.47	C577087580	2,472,250.47	0.00	C660528098	0.00	0.00	1	0

Figure 2.1: Sample of the PaySim Dataset

The original dataset consists of 6,354,407 legitimate transactions and 8,213 fraudulent transactions, making it highly imbalanced. To reduce the bias toward the majority class, legitimate transactions were downsampled. The final dataset contains 41,065 legitimate

and 8,213 fraudulent transactions, maintaining an approximate 5:1 class ratio.

2.2 Data Preprocessing

The preprocessing pipeline consists of data cleaning, feature engineering, feature scaling, and dataset splitting.

Invalid records were removed, numerical attributes were converted into suitable formats, and only `TRANSFER` and `CASH_OUT` transactions were retained. Additional features, including log-transformed transaction amount, sender and receiver balance differences, transaction hour, night transaction indicator, and high-value transaction indicator, were created to improve model performance. The transaction type was numerically encoded before training.

All numerical features were standardized using the training data. The dataset was then divided into training, validation, and testing sets while preserving the fraud-to-legitimate ratio. Class weighting was also applied during training to improve learning from the minority fraud class.

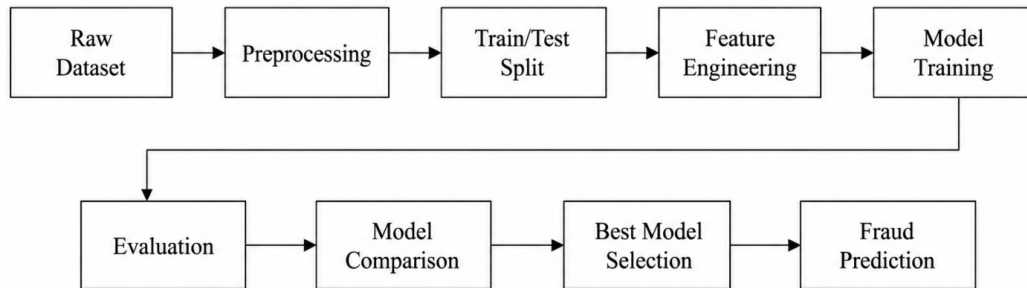


Figure 2.2: Overall Fraud Detection Pipeline

2.3 Models Used

Six deep learning models were implemented and evaluated using the same preprocessed dataset.

- Multilayer Perceptron (MLP)

- Bidirectional Long Short-Term Memory (BiLSTM)
- Gated Recurrent Unit (GRU)
- TabNet
- FT-Transformer
- TabTransformer

The MLP model was used as the baseline neural network. GRU and BiLSTM were employed to capture sequential transaction patterns, while TabNet, FT-Transformer, and TabTransformer were selected because they are specifically designed for learning from tabular data.

2.4 Evaluation Metrics

The performance of the developed models was evaluated using several standard classification metrics.

- **Confusion Matrix:** Shows how correctly and incorrectly the model classifies fraudulent and legitimate transactions.
- **Precision:** Measures the proportion of transactions predicted as fraudulent that are actually fraudulent.
- **Recall:** Measures the proportion of actual fraudulent transactions that are correctly identified by the model.
- **F1-Score:** Represents the harmonic mean of Precision and Recall, providing a balanced measure of fraud detection performance.
- **ROC-AUC:** Measures the model's ability to distinguish between fraudulent and legitimate transactions across different classification thresholds.
- **PR-AUC:** Measures the area under the Precision-Recall curve and is particularly useful for evaluating models on highly imbalanced datasets.
- **Log Loss:** Measures the difference between predicted probabilities and the actual class labels. Lower values indicate better model performance.

Chapter 3

Results & Discussion

3.1 Overall Model Performance

Six deep learning models were evaluated using the same preprocessed dataset and experimental settings. Their performance was compared using Precision, Recall, F1-Score, ROC-AUC, PR-AUC, and Log Loss. Table 3.1 summarizes the performance of all evaluated models.

Table 3.1: Performance Comparison of Deep Learning Models

Model	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Log Loss
MLP	0.968900	0.986201	0.977474	0.999058	0.996666	0.028745
BiLSTM	0.977556	0.954545	0.965914	0.997712	0.995123	0.052937
GRU	0.972750	0.956169	0.964388	0.997728	0.994621	0.059824
TabNet	0.916201	0.931818	0.923944	0.995199	0.978575	0.103618
TabTransformer	0.864143	0.944805	0.902675	0.992234	0.964021	0.097173
FT-Transformer	0.778118	0.825487	0.801103	0.976149	0.896989	0.185497

Among all evaluated models, the Multilayer Perceptron (MLP) achieved the best overall performance. It obtained the highest Recall (0.986201), F1-Score (0.977474), ROC-AUC (0.999058), and PR-AUC (0.996666), while also producing the lowest Log Loss (0.028745). These results indicate that MLP provided the most reliable fraud detection performance.

3.2 F1-Score Comparison

Figure 3.1 compares the fraud-class F1-Score achieved by all models. MLP obtained the highest F1-Score, indicating the best balance between Precision and Recall for fraud detection.

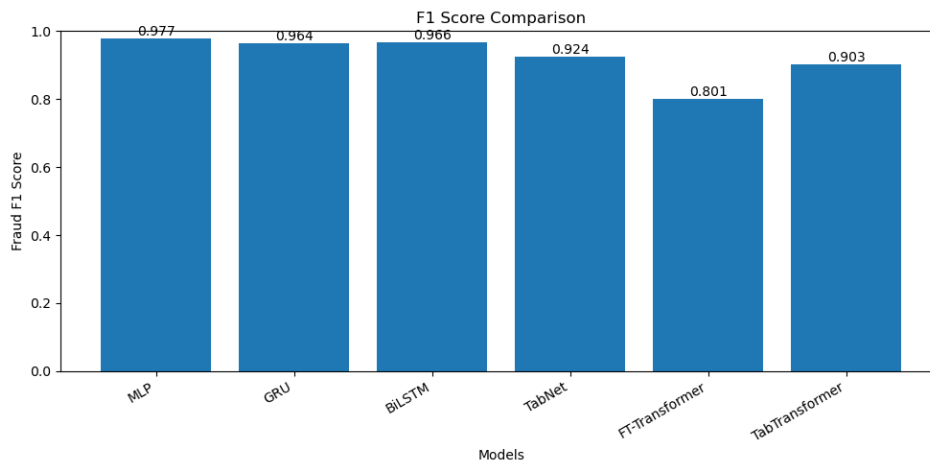


Figure 3.1: Fraud-Class F1-Score Comparison

3.3 Confusion Matrix Analysis

The confusion matrix of the best-performing MLP model is shown in Figure 3.2. The model correctly classified most legitimate and fraudulent transactions while producing only a small number of false positives and false negatives, demonstrating reliable classification performance.

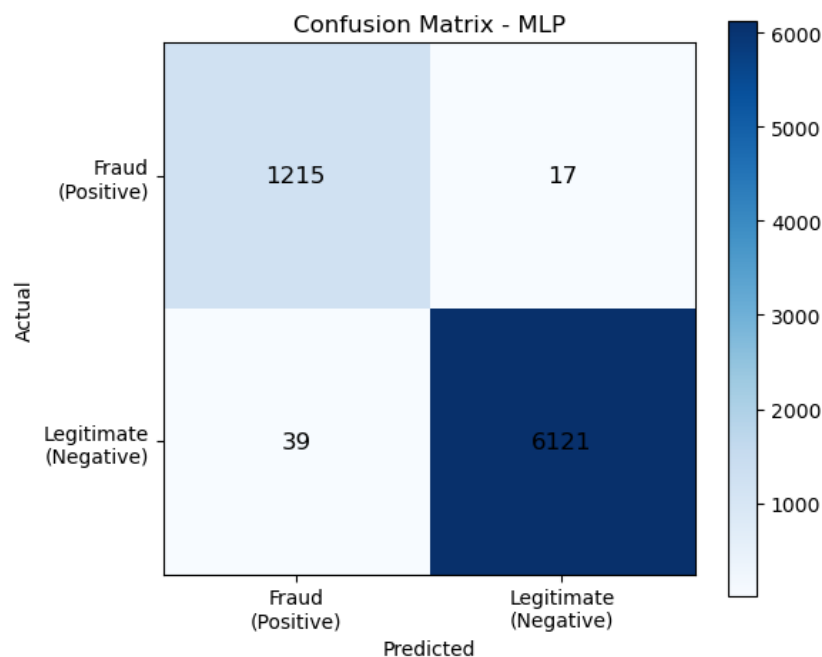


Figure 3.2: Confusion Matrix of the MLP Model

3.4 False Positive and False Negative Analysis

False positives and false negatives are critical in fraud detection because they represent false alarms and missed fraud cases. As shown in Figure 3.3, MLP produced the fewest false negatives while maintaining a low number of false positives compared to the other models.

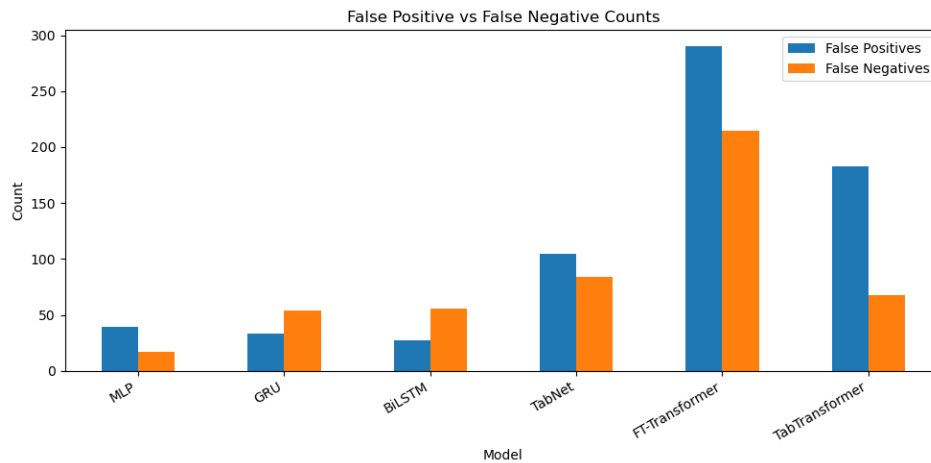


Figure 3.3: False Positive and False Negative Comparison

3.5 Fraud Detection Metrics

Figure 3.4 compares the fraud-class Precision, Recall, and F1-Score of all evaluated models. MLP achieved the highest Recall and F1-Score, demonstrating its ability to identify fraudulent transactions while maintaining strong overall performance.

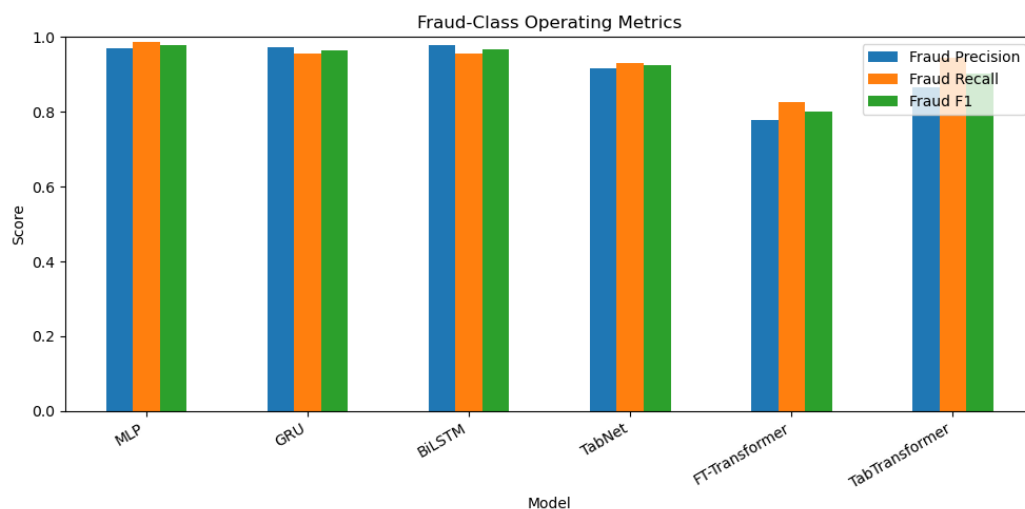


Figure 3.4: Fraud-Class Precision, Recall, and F1-Score

3.6 PR-AUC Analysis

The PR-AUC metric is particularly important for highly imbalanced datasets. As illustrated in Figure 3.5, MLP achieved the highest PR-AUC among all evaluated models. The Precision-Recall curves shown in Figure 3.6 further confirm its superior fraud detection capability.

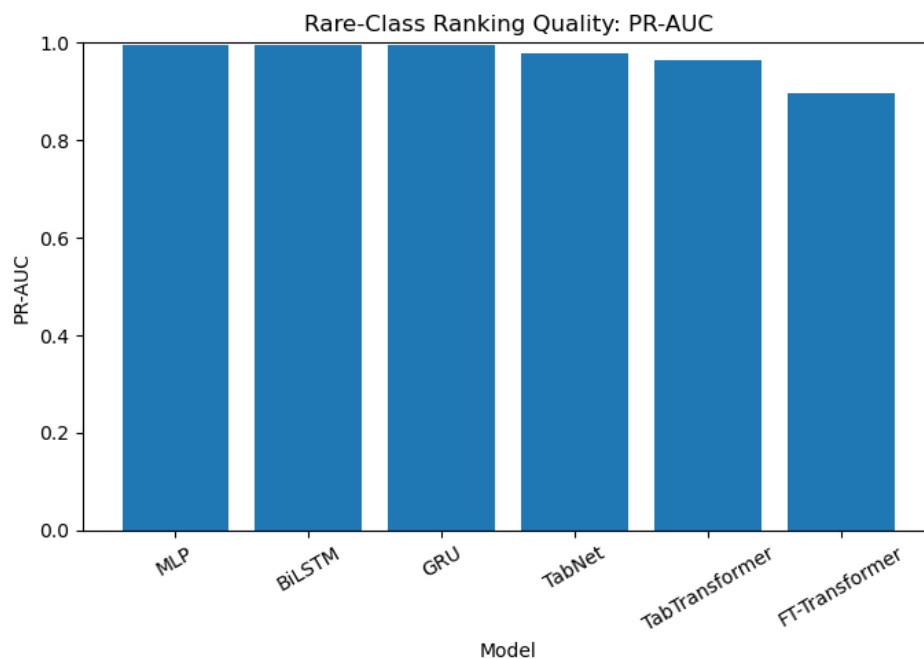


Figure 3.5: PR-AUC Comparison

3.7 ROC-AUC Analysis

Figure 3.7 presents the ROC curves of all evaluated models. MLP achieved the highest ROC-AUC score of 0.999058, indicating excellent discrimination between fraudulent and legitimate transactions.

3.8 Training and Validation Loss

The training and validation loss curves of the MLP model are shown in Figure 3.8. Both curves decrease steadily during training without noticeable divergence, indicating stable learning and no significant overfitting.

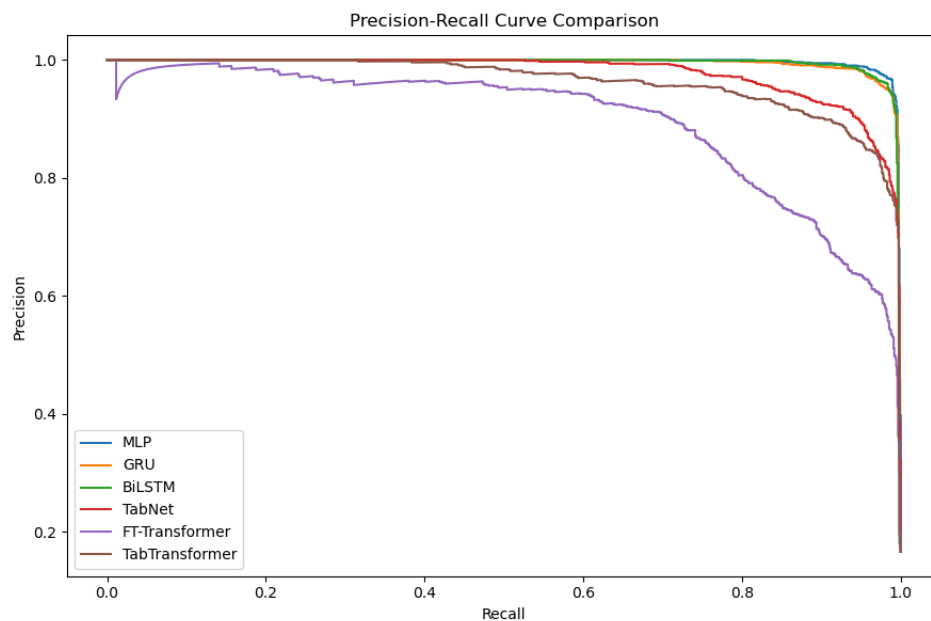


Figure 3.6: Precision-Recall Curves

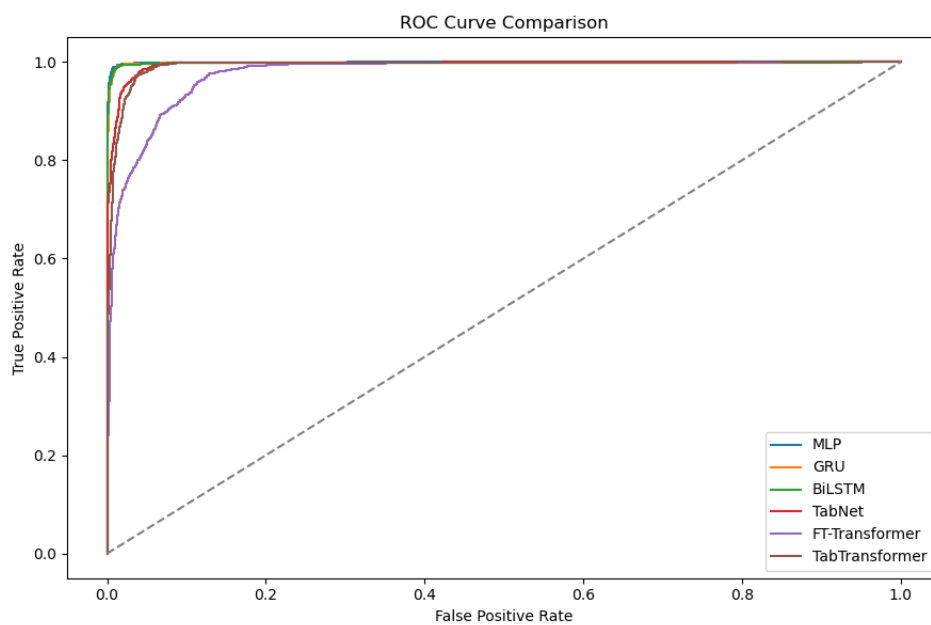


Figure 3.7: ROC Curve Comparison

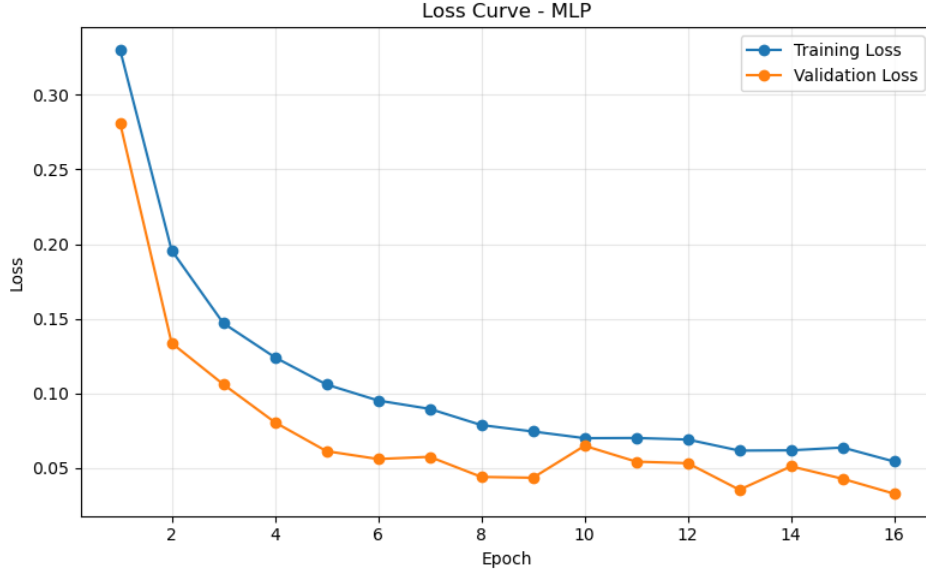


Figure 3.8: Training and Validation Loss of the MLP Model

3.9 Discussion

Based on the experimental results, the MLP model was selected as the best-performing model for the proposed fraud detection framework. It consistently outperformed the other deep learning models across all major evaluation metrics while maintaining the lowest Log Loss.

The strong performance of MLP can be attributed to the engineered transaction-level features, including transaction amount, balance differences, transaction type, and time-related features. These features effectively captured the characteristics of fraudulent transactions without requiring complex sequential modeling.

Although GRU and BiLSTM also achieved competitive results, the transaction records in this dataset do not represent sequential user histories, limiting the advantage of sequence-based architectures. Similarly, TabNet, FT-Transformer, and TabTransformer performed well but did not surpass the MLP model.

Overall, the experimental results demonstrate that the proposed MLP-based framework provides an accurate, robust, and reliable solution for detecting fraudulent financial transactions in highly imbalanced datasets.

Chapter 4

Streamlit Application

4.1 System Workflow

A Streamlit-based web application was developed to demonstrate the proposed fraud detection framework in an interactive environment. Users enter the required transaction details through a simple interface, after which the application automatically performs data preprocessing, feature engineering, feature scaling, and fraud prediction using the trained MLP model. The prediction result is then displayed together with the corresponding fraud probability.

4.2 Application Features

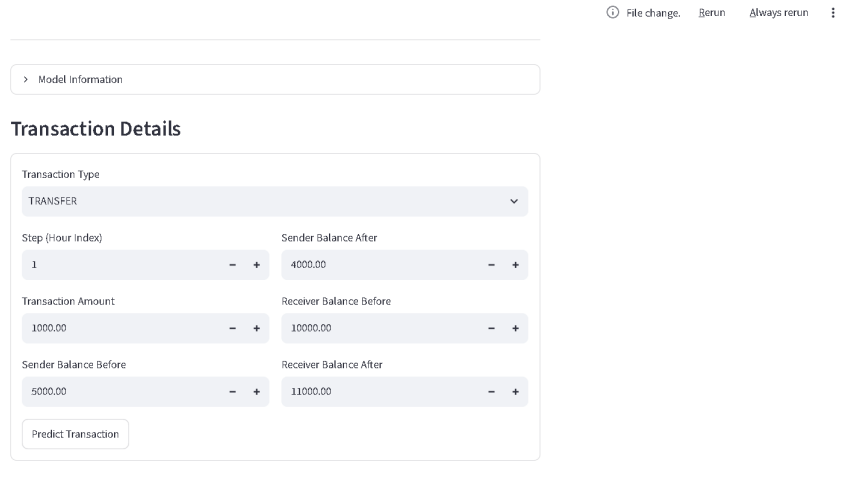
The application provides the following features:

- Simple and user-friendly transaction input interface.
- Automatic preprocessing and feature engineering before prediction.
- Real-time fraud prediction using the trained MLP model.
- Displays fraud probability and prediction result.
- Supports prediction for both legitimate and fraudulent transactions.

4.3 Application Screenshots

4.3.1 Transaction Input Form

The transaction input form allows users to enter the required transaction details, including transaction type, amount, and sender and receiver account balances. After entering the information, users can submit the transaction for fraud prediction.

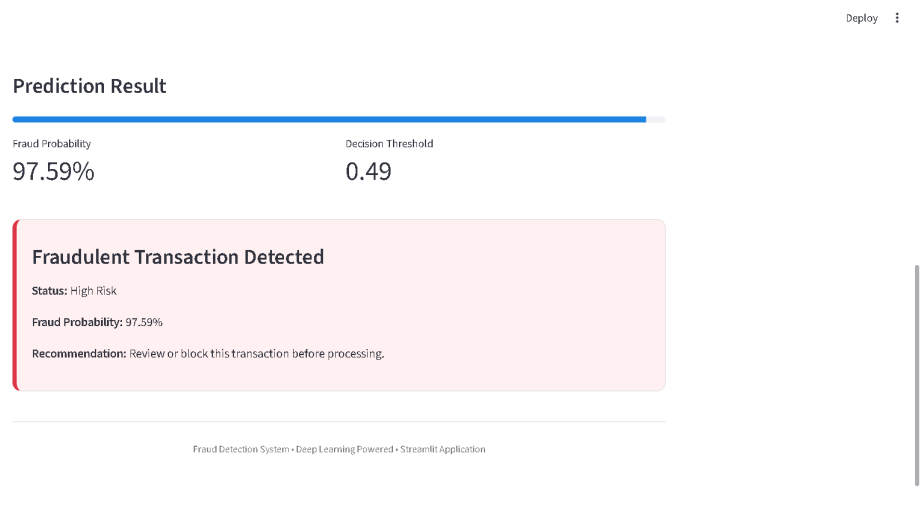


The screenshot shows a web application interface for entering transaction details. At the top right, there are status indicators: 'File change', 'Run', 'Always rerun', and a settings icon. Below these is a tab labeled 'Model Information'. The main section is titled 'Transaction Details' and contains several input fields with numerical values and minus/plus buttons for adjustment. The fields are: Transaction Type (dropdown menu set to 'TRANSFER'), Step (Hour Index) (value 1), Sender Balance After (value 4000.00), Transaction Amount (value 1000.00), Receiver Balance Before (value 10000.00), Sender Balance Before (value 5000.00), and Receiver Balance After (value 11000.00). At the bottom of the form is a button labeled 'Predict Transaction'.

Figure 4.1: Transaction Input Form

4.3.2 Fraudulent Transaction Prediction

When a transaction is classified as fraudulent, the application displays a warning message together with the predicted fraud probability, helping users identify suspicious transactions.



The screenshot displays the 'Prediction Result' section of the application. It features a progress bar at the top. Below the progress bar, there are two columns: 'Fraud Probability' showing '97.59%' and 'Decision Threshold' showing '0.49'. A red-bordered box contains the following information: 'Fraudulent Transaction Detected', 'Status: High Risk', 'Fraud Probability: 97.59%', and 'Recommendation: Review or block this transaction before processing.' At the bottom of the page, there is a footer text: 'Fraud Detection System • Deep Learning Powered • Streamlit Application'.

Figure 4.2: Fraudulent Transaction Prediction

4.3.3 Legitimate Transaction Prediction

When a transaction is classified as legitimate, the application displays a confirmation message along with the predicted fraud probability, indicating that the transaction appears to be safe.

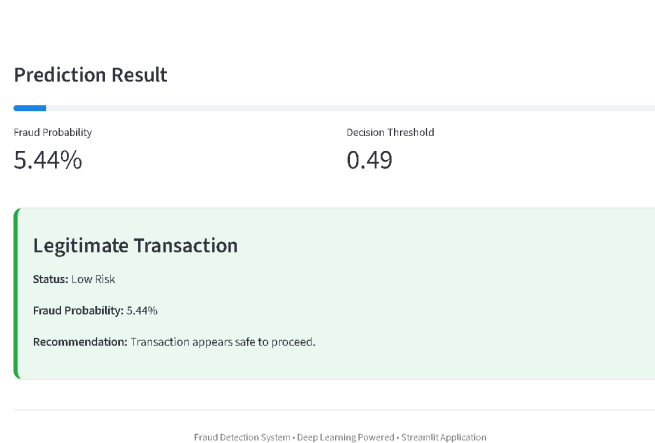


Figure 4.3: Legitimate Transaction Prediction

Chapter 5

Conclusion & Future Work

5.1 Conclusion

This project presented a deep learning-based framework for detecting fraudulent financial transactions using the PaySim dataset. After data preprocessing, feature engineering, and handling class imbalance, six deep learning models were developed and evaluated under the same experimental settings. Their performance was compared using Precision, Recall, F1-Score, ROC-AUC, PR-AUC, and Log Loss.

Among the evaluated models, the Multilayer Perceptron (MLP) achieved the best overall performance, obtaining the highest F1-Score, ROC-AUC, and PR-AUC while maintaining the lowest Log Loss. The results demonstrate that the proposed framework can accurately distinguish fraudulent transactions from legitimate ones and provide reliable performance on highly imbalanced financial transaction data.

A Streamlit-based web application was also developed to demonstrate the practical use of the trained model, enabling users to perform fraud prediction through a simple and interactive interface.

5.2 Future Work

Although the proposed framework achieved promising results, several improvements can be explored in future research:

- Integrate the model with real-time banking transaction systems.
- Evaluate the framework using real-world financial datasets from multiple institutions.

- Investigate advanced transformer-based architectures and ensemble learning methods.
- Apply automated hyperparameter optimization to further improve model performance.
- Develop explainable AI techniques to provide interpretable fraud prediction decisions.
- Extend the system to support real-time fraud monitoring and alert generation.
- Deploy the application as a scalable cloud-based fraud detection service.

Acknowledgments

Project Contributor

- **Srabonti Suchi Talukdar**
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AI Assistance Tools

- **ChatGPT (OpenAI):** Project planning, technical guidance, debugging, documentation support, and content refinement.
- **Google Gemini:** Technical assistance, research support, and code-related guidance.

Technologies Used

- Python
- TensorFlow
- PyTorch
- Streamlit